# 更换 Cisco Nexus 3232C 存储交换机

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# 更换 Cisco Nexus 3232C 存储交换机

更换 Cisco Nexus 3232C 存储交换机时，您必须了解某些配置信息，端口连接和布线要求。

在 Cisco Nexus 存储交换机上安装 NX-OS 软件和 RCF 之前，必须验证是否存在以下条件：

- 您的系统可以支持 Cisco Nexus 3232C 存储交换机。
- 您必须已查看 Cisco 以太网交换机页面上的交换机兼容性表以了解受支持的 ONTAP ， NX-OS 和 RCF 版本。

> 您应了解 RCF 和 NX-OS 版本中的命令语法之间可能存在依赖关系。

- 您必须已参考 Cisco 网站上提供的相应软件和升级指南，以获取有关 Cisco 交换机升级和降级过程的完整文档 "Cisco Nexus 3000 系列交换机"。
- 您必须已下载适用的 RCF 。

# 更换 Cisco Nexus 3232C 存储交换机的步骤

您可以通过执行一系列特定任务无中断地更换有故障的 Cisco Nexus 3232C 存储交换机。

现有网络配置必须具有以下特征：

- Cisco 以太网交换机页面会为您的交换机提供最新的 RCF 和 NX-OS 版本。
- 两台交换机上必须存在管理连接。

> 确保已完成所有故障排除步骤，以确认您的交换机需要更换。

更换用的 Cisco Nexus 3232C 交换机必须具有以下特征：

- 管理网络连接必须正常工作。
- 必须能够通过控制台访问更换用的交换机。
- 必须将相应的 RCF 和 NX-OS 操作系统映像加载到交换机上。
- 必须完成交换机的初始自定义。
- 操作步骤 摘要： *
- * 确认要更换的交换机为 S2 （步骤 1.-5.*
- * 断开交换机 S2 的缆线 （第 6 步）*
- * 将缆线重新连接到交换机 NS2 （步骤 7.*
- * 验证交换机 NS2 上的所有设备配置（步骤 8.-10*
- 此操作步骤 将第二个 Nexus 3232C 存储交换机 S2 替换为新的 3232C 交换机 NS2 。
- 这两个节点分别为 node1 和 node2 。

步骤

1. 【一个】如果此集群上启用了 AutoSupport ，则通过调用 AutoSupport 消息禁止自动创建案例： `ssystem node AutoSupport invoke -node * -type all - message MAIN=xh`

   *x* 是维护时段的持续时间，以小时为单位。

   > AutoSupport 消息会通知技术支持此维护任务，以便在维护窗口期间禁止自动创建案例。

2. 检查存储节点端口的运行状况，确保已连接到存储交换机 S1： `storage port show -port-type ENET`

```
storage::*> storage port show -port-type ENET
                                          Speed                      VLAN
Node                Port Type  Mode     (Gb/s) State    Status       ID
------------------ ---- ----- ------- ------ -------- --------- ----
node1
                    e3a  ENET  storage    100 enabled  online       30
                    e3b  ENET  storage      0 enabled  offline      30
                    e7a  ENET  storage      0 enabled  offline      30
                    e7b  ENET  storage      0 enabled  offline      30
node2
                    e3a  ENET  storage    100 enabled  online       30
                    e3b  ENET  storage      0 enabled  offline      30
                    e7a  ENET  storage      0 enabled  offline      30
                    e7b  ENET  storage      0 enabled  offline      30
```

3. 验证存储交换机 S1 是否可用： `network device-discovery show`

```
storage::*> network device-discovery show
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- -----------------
----------------
node1/cdp
            e3a    S1                        Ethernet1/1       NX3232C
            e4a    node2                     e4a               AFF-A700
            e4e    node2                     e4e               AFF-A700
node1/lldp
            e3a    S1                        Ethernet1/1       -
            e4a    node2                     e4a               -
            e4e    node2                     e4e               -
node2/cdp
            e3a    S1                        Ethernet1/2       NX3232C
            e4a    node1                     e4a               AFF-A700
            e4e    node1                     e4e               AFF-A700
node2/lldp
            e3a    S1                        Ethernet1/2       -
            e4a    node1                     e4a               -
            e4e    node1                     e4e               -
```

4. 对工作交换机运行 `sHow LLDP neighbors` 命令，确认您可以同时看到节点和所有磁盘架：`sHow LLDP neighbors`

```
S1# show lldp neighbors
Capability codes:
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
Device ID               Local Intf      Hold-time  Capability  Port ID
node1                   Eth1/1          121        S           e3a
node2                   Eth1/2          121        S           e3a
SHFGD2008000011         Eth1/5          121        S           e0a
SHFGD2008000011         Eth1/6          120        S           e0a
SHFGD2008000022         Eth1/7          120        S           e0a
SHFGD2008000022         Eth1/8          120        S           e0a
```

5. 【五】验证存储系统中的磁盘架端口：`storage shelf port show -fields remote-device ,remote-port`

```
storage::*> storage shelf port show -fields remote-device,remote-port

shelf  id  remote-port  remote-device
-----  --  -----------  -------------
3.20   0   Ethernet1/5  S1
3.20   1   -            -
3.20   2   Ethernet1/6  S1
3.20   3   -            -
3.30   0   Ethernet1/7  S1
3.20   1   -            -
3.30   2   Ethernet1/8  S1
3.20   3   -            -
```

6. 【六个】拔下连接到存储交换机 S2 的所有缆线。
7. 【七条】将所有缆线重新连接到更换用的交换机 NS2 。
8. 【八个】重新检查存储节点端口的运行状况： `storage port show -port-type ENET`

```
storage::*> storage port show -port-type ENET
                                         Speed                        VLAN
Node                Port Type  Mode     (Gb/s) State    Status         ID
------------------ ---- ----- ------- ------ -------- ------------ ---
node1
                    e3a  ENET  storage    100 enabled  online        30
                    e3b  ENET  storage      0 enabled  offline       30
                    e7a  ENET  storage      0 enabled  offline       30
                    e7b  ENET  storage    100 enabled  online        30
node2
                    e3a  ENET  storage    100 enabled  online        30
                    e3b  ENET  storage      0 enabled  offline       30
                    e7a  ENET  storage      0 enabled  offline       30
                    e7b  ENET  storage    100 enabled  online        30
```

9. 验证两个交换机是否均可用： `network device-discovery show`

```
storage::*> network device-discovery show
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------  --------
node1/cdp
            e3a    S1                        Ethernet1/1       NX3232C
            e4a    node2                     e4a               AFF-A700
            e4e    node2                     e4e               AFF-A700
            e7b    NS2                       Ethernet1/1       NX3232C
node1/lldp
            e3a    S1                        Ethernet1/1       -
            e4a    node2                     e4a               -
            e4e    node2                     e4e               -
            e7b    NS2                       Ethernet1/1       -
node2/cdp
            e3a    S1                        Ethernet1/2       NX3232C
            e4a    node1                     e4a               AFF-A700
            e4e    node1                     e4e               AFF-A700
            e7b    NS2                       Ethernet1/2       NX3232C
node2/lldp
            e3a    S1                        Ethernet1/2       -
            e4a    node1                     e4a               -
            e4e    node1                     e4e               -
            e7b    NS2                       Ethernet1/2       -
```

10. 【十】验证存储系统中的磁盘架端口： `storage shelf port show -fields remote-device , remote-port`

```
storage::*> storage shelf port show -fields remote-device,remote-port
shelf id remote-port remote-device
----- -- ----------- -------------
3.20  0  Ethernet1/5 S1
3.20  1  Ethernet1/5 NS2
3.20  2  Ethernet1/6 S1
3.20  3  Ethernet1/6 NS2
3.30  0  Ethernet1/7 S1
3.20  1  Ethernet1/7 NS2
3.30  2  Ethernet1/8 S1
3.20  3  Ethernet1/8 NS2
```

11. 如果禁止自动创建案例，请通过调用 AutoSupport 消息重新启用此功能： `ssystem node AutoSupport invoke -node * -type all -message MAINT=end`

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